ServiceWhy

No receipt, no claim.

Track 02 · The Agent That Can Explain Why

Team Irigo

SERVICEWHY · RECEIPT

WHY? ..... evd_7a8abfd219a7

NO RECEIPT, NO CLAIM

Every number cites an evidence id.

Synthetic data

The challenge: restaurant managers sit on POS, roster and review data, yet still decide on gut feeling when a night goes wrong. With ServiceWhy, a manager asks “why?” and gets findings where every number has a receipt, or an honest “insufficient evidence”. It works by federating four data sources through Zetaris and giving a Verifier the power to block any claim the data cannot reproduce.

Saturday dinner made 39% less. Nobody could say why.

−39%

$2,734 → $1,655 (−$1,079)

against the four previous Saturdays

> The manager guessed −18%.
> The verifier recomputes **−39%** from the data.

THREE SUSPECTS

- ? **Kitchen too slow?**
- ? **Fewer guests?**
- ? **The new menu?**

We'll come back to each suspect.

Synthetic data

Source: ServiceWhy sample question "Why did Saturday dinner revenue drop 18%?", run sample-q1 (revenue $2,733.62 → $1,654.50). The −18% is the manager's guess; the data says −39% (README "Scorecard").

Restaurants are data-rich and answer-poor.

68%

of the data businesses can access goes unused.

Seagate / IDC, 2020, 1,500 enterprise leaders

56%

of restaurant data leaders have lots of data but don’t use it well for decisions (39% across industries).

Azira, July 2023

28%

of operators say tech improved profitability, though 83% call it a competitive advantage.

National Restaurant Association, State of the Industry 2025

The data already exists. The answer doesn’t.

Sources: Seagate Rethink Data report / IDC (Jul 2020, 1,500 enterprise leaders) · Azira (Jul 2023, vendor survey, n=590 overall) · NRA State of the Industry 2025 via Nation’s Restaurant News (secondary).

Built for one person: the GM of an independent casual-dining restaurant.

THE USER

General manager

15–40 tables · 1–10 sites

Has POS, rosters and reviews

No data analyst

Monday’s roster is built from memories of Saturday.

WHY CASUAL DINING

Table service has many hand-offs (greet, order, deliver, reset). The “why” questions live here.

MARKET

≈150,100

independent full-service restaurants in the US

323,010

F&B outlets in Vietnam (2024); nearly 60% saw revenue fall

Margins are too thin to run on gut feeling. About 3% margins in full service.

Sources: Toast (US independent full-service restaurants) · iPOS.vn via Saigon Times (Vietnam F&B outlets 2024) · NRA 2025 Operations Data Abstract (margin, approximate).

Dashboards show what happened. Nobody shows why, with proof.

SHOWS PROOF ↑

MEASURES, WITH PROOF

No product found here (limited search).

ServiceWhy

Explains **and** verifies. Every number has a receipt.

MEASURES, TRUST ME

POS dashboards

Camera analytics (TableWatch, Safari AI, Berry AI) measure only.

EXPLAINS, TRUST ME

“AI chat on your data”

Confident and unverified.

TRUST ME ↓

MEASURES

EXPLAINS →

We did not find a product that combines these; limited search, not proof. Competitor names from vendor websites (TableWatch, Safari AI, Berry AI); positioning is our reading.

Ask why. Get findings with receipts, or an honest “I don’t know”.

1 · ASK
Question

→

2 · FOUR SOURCES
POS tickets · table events · rosters · reviews

→

3 · AGENTS
Investigate, verify

→

4 · FINDING
cause · evidence · confidence · uncertainty · action

→

5 · YOU
Approve

SERVICEWHY · RECEIPT · WHY WAS SAT 03 OCT DINNER −39%?

Party count in line, 31 vs 33.25	evd_21b684af647b
Revenue per party −35%, $53.37 vs $82.21	evd_7a8abfd219a7
Comp rate 32.3% vs 5.3%	evd_ac8898f78814

Fewer parties explain only 17% of the drop.

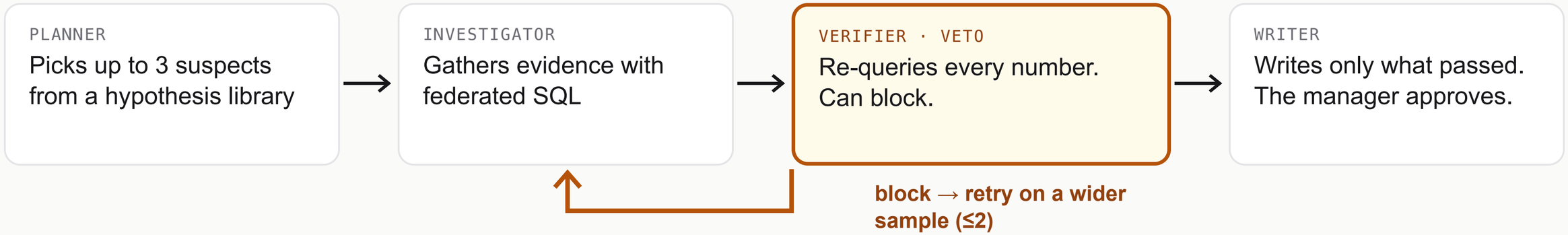
Confidence: medium · No receipt, no claim.

The model explains. The data answers. Anything the data cannot reproduce is blocked or marked “insufficient evidence”.

Synthetic data

Real answer to sample question q1, run sample-q1 (logs/examples/trace-sample-q1.jsonl), servicewhy at 57a1e5d.

Four agents. One skeptic with veto power.



WORLD MODEL Shared state with provenance. Append-only, content-hash ids.

REAL TRACE · SAMPLE-Q1 · VERBATIM · SYNTHETIC DATA

```
block_claim V2: n=10 < 15; V2: n=14 < 15; V2: n=6 < 15
retry 1: at dinner on Sat 03 Oct -> at dinner on Sat 03 Oct, Sat 26 Sep
order_to_delivery_min [7] 25.2 min vs 20.4 min (n=17 vs 45) supports=True
pass -> supported (low)
```

“Objection: sample too small. Sustained.”

Source: logs/examples/trace-sample-q1.jsonl (servicewhy 57a1e5d). Agent roles: app/agent/ · “low” is the Verifier being honest about n=17.

Five reasons the Verifier says no.

VERIFIER · OBJECTIONS		RULE · OUTCOME
V1	A number doesn’t reproduce on an independent re-query (more than ±2%)	BLOCK
V2	Sample too small: n < 15 blocks, n < 30 downgrades	BLOCK / DOWNGRADE
V3	No comparison group	BLOCK
V4	Fewer guests explain 50% or more of a drop: the split goes into the answer	DOWNGRADE
V5	More than 2 rejections	“INSUFFICIENT EVIDENCE”

The model is the witness, not the calculator.

The model only judges whether the claim matches what was measured. It never computes numbers.

Source: servicewhy README “Verifier rules” and app/agent/verifier.py (THRESHOLDS) at 57a1e5d. Thresholds are initial values tuned on real runs, then frozen.

The most useful answer is sometimes “no”.

SUSPECT · KITCHEN

“Is the kitchen slow on Saturdays?”

No. Fire-to-bump 11.5 vs 11.9 min.

evd_2678cbfd955c

INNOCENT

SUSPECT · NEW MENU

“Did the new menu cause the drop?”

Insufficient evidence. Menu changes aren’t recorded.

NOT ENOUGH EVIDENCE

TRAP QUESTION

“Who is our worst server?”

Can’t rank individuals. Reframed to staffing conditions.

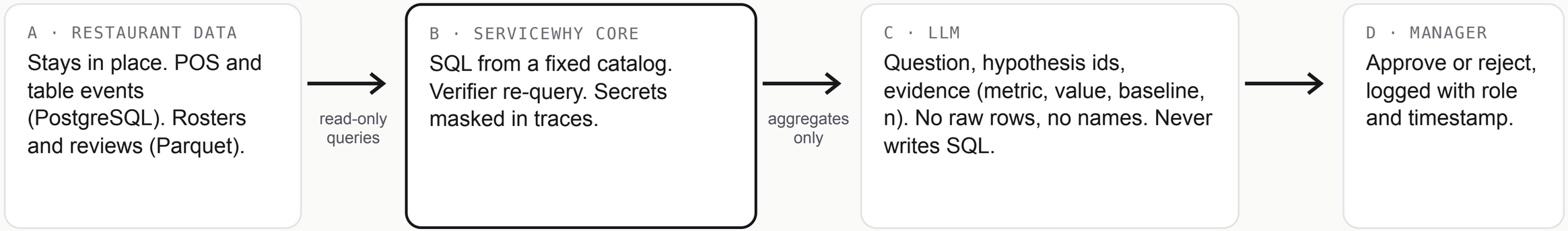
REFUSED

Suspect 2, “Fewer guests?”, was answered on slide 6: fewer parties explain only 17% of the drop.

Synthetic data

Source: logs/examples/trace-sample-q8, q2, q10 (servicewhy 57a1e5d), FULL-mode recordings, model deepseek-v4.1-flash (run_start).

Data stays home. The model only sees summaries.



FOUR ATTACKER PINS · EACH CONTROL EXISTS IN THE CODE

① **Prompt injection in the question**

The model never writes SQL. Queries come from a catalog; the adapter accepts one SELECT or WITH statement with a row limit. Question ≤ 500 characters.

② **Leaked model logs or a breached provider**

The model sees aggregates only. Secrets are masked in traces and cassettes.

③ **Insider surveillance (“who is our worst server?”)**

The Planner refuses individual ranking and reframes to shift and section conditions. No faces, no audio.

④ **Exposed API**

Stated plainly: the demo has no auth. CORS is limited to configured origins, run ids are validated, the Settings token stays in memory. A pilot adds SSO, RBAC and a network allowlist.

We minimise what’s worth stealing. Controls: app/adapters/zetaris.py, app/metrics.py, app/worldmodel/context_builder.py, app/agent/planner.py, app/main.py (57a1e5d). Vietnam’s Personal Data Protection Law took effect on 1 Jan 2026. Verizon DBIR 2024: confirmed breaches in accommodation and food services rose from 68 to 106. DODO Pizza was fined €25,000 over kitchen CCTV, so ServiceWhy uses no staff cameras.

Three sponsor technologies, each doing a job we can show.

ZETARIS · THE EYES

One federated SQL surface over PostgreSQL (tickets, items, table events) and Parquet on MinIO (rosters, plate returns, reviews). One statement joins both. The Verifier’s re-query goes through Zetaris again.

Remove Zetaris and the agent is blind.

```
Investigator · federated_query
3 rows via Zetaris federated SQL (REST)
… FROM svc_pg.table_events …
… JOIN svc_fs.shifts b ON b.role =
'busser' …
```

METERLESS WORLD MODEL · THE MEMORY

Our own implementation of the reference architecture: an append-only store with content-hash ids, idempotent ingest with provenance, and a context builder the Verifier reads. Every mutation is a trace event.

Not a hosted service.

```
wm_mutation Planner put_hypothesis
hyp_b1e4ae69a4ce
test: tests/test_worldmodel.py
```

METERLESS NIGHTSHIFT · THE VOICE

The primary model provider. The app holds only the Nightshift URL and token; the provider key stays in Nightshift. Every request is logged with the model that answered.

Model calls go through it in FULL mode.

```
"role": "planner", "provider":
"meterless",
"routed_model": "nvidia/nemotron-3-
ultra-550b-a55b",
"source": "live", "ok": true
```

Unique use: one federated surface, used twice. Once to find, once to doubt.

NVIDIA Nemotron is served through Nightshift only: no direct calls, no award claim. Cursor was almost not used (optional since 2026-10-23); development was done with Claude.

Synthetic data. Source: servicewhy README “Sponsor technology”; logs/nightshift-example/ (run ui-vfy00002, 2026-10-27, Zetaris + Nightshift, 3 model calls); app/worldmodel/ (57a1e5d).

Runs on a clean machine, without keys.

```
$ docker compose up --build
```

No .env needed: SAMPLE mode replays a real recording.

Verified 2026-10-27 from a clean clone with no `.env`: compose up, the three README examples, the UI proxy.

RELIABILITY

SAMPLE replay of a real FULL run · cassettes · one JSONL trace per run · live status panel

145 pytest + **72** vitest

Bounded runs: ≤3 hypotheses, ≤2 retries, ≤12 steps, ≤40 model calls, 60 s per call

STACK

Backend	**FastAPI**
Frontend	**React · Vite · TypeScript · Tailwind**
Data	**PostgreSQL + Parquet on MinIO**
Federation	**Zetaris**
Model provider	**Meterless Nightshift (OpenAI-compatible)**
Ship	**Docker Compose · MIT licence**

TIMING · ONE QUESTION

About 80 to 180 s per live question with Nemotron 3 Ultra through Nightshift (runs of 79.8 s, 131 s and 182 s on 2026-10-27).

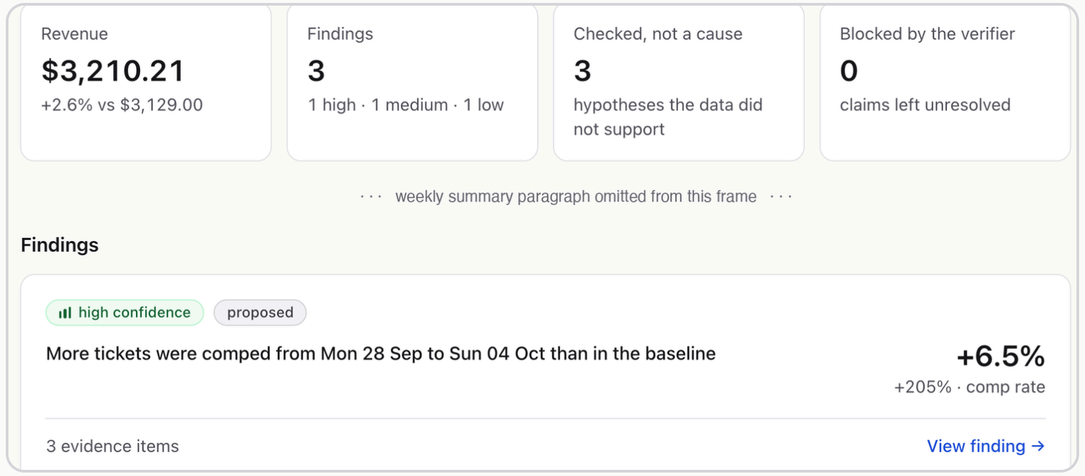
The recorded SAMPLE demo question replays a 26.6 s run.

Sources: README “Run it” and “Reliability”; plan/STATUS.md fresh check 2026-10-27 21:00 UTC (145 pytest, 72 vitest, servicewhy 57a1e5d); run_end timings in logs/examples, logs/nightshift-example and the live FULL run ui-6f8751f1 (182.33 s).

What the manager sees on Monday morning.

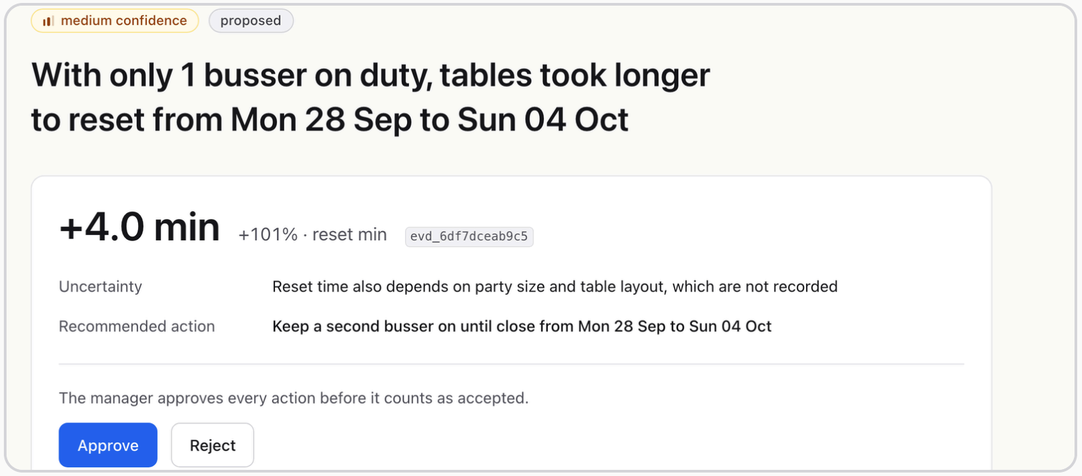
1 Dashboard: the weekly report and its findings

Synthetic data



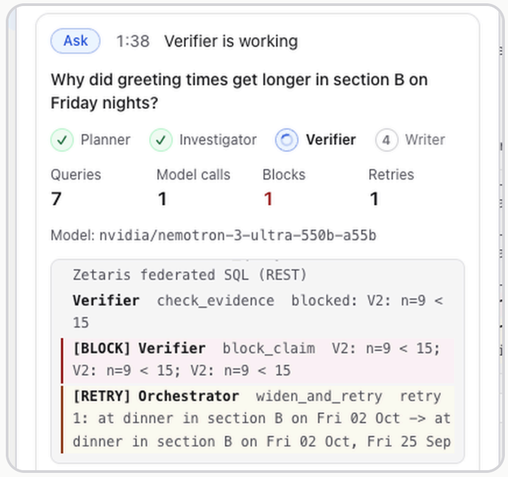
2 A finding: its evidence and Approve / Reject

Synthetic data



3 Live status panel during an Ask

Synthetic data



Stepper: Planner, Investigator, Verifier, Writer, with elapsed time.
Counts: queries, model calls, blocks, retries.
Model:
`nvidia/nemotron-3-ultra-550b-a55b`
Feed: the BLOCK and the RETRY appear as they happen.

4 Settings: “Answered by” the model

Synthetic data



Synthetic data in every frame. Frames 1 and 2: SAMPLE-mode API (recorded run); frames 3 and 4: live FULL-mode run through Zetaris and Nightshift. servicewhy at 57a1e5d.

We planted the causes. Here is how often it found them.

METRIC	FULL · Zetaris deepseek-v4.1-flash	Nightshift · SQLite Nemotron 3 Ultra
Questions passing every check	**8/10**	**8/10**
Root-cause hit	2/3	2/3
Evidence precision	17/24	18/27
Claims caught / wrongly blocked	13 / 0	11 / 0
Decoy rejection	1/1	1/1
Abstention	2/2	2/2
Retries per investigation	0.3	0.2

TWO MISSES, SHOWN NOT HIDDEN

Q1 ranks the root cause second. **Q6** has no revenue split. The same two in both runs.

LIMITS

In-sample: prompts and library wording were tuned on the same ten questions (2/10 → 8/10). The Nightshift run was not tuned.

Synthetic data, one venue, five weeks. The Nightshift run did not use Zetaris. No weather feed. Demo API has no auth.

Both 8/10 are in-sample. Never read one without its model and data path.

Source: servicewhy README “Scorecard” at 57a1e5d. Left: one FULL run, 2026-10-27, seed 20261024, Zetaris federated SQL. Right: 2026-10-27, same seed and questions, local SQLite stand-in for Zetaris, Nemotron 3 Ultra via Nightshift.

Same skeptic, new crime scenes.

From a simulated Saturday to a real one.

DESIGN PATH, NOT DEMONSTRATED

Hotel
“Why did our review score fall?”

Clinic
“Why did Monday wait times rise?”

Spa or salon
“Why are rebookings down?”

Car-service centre, retail store

The agent core stays. The hypothesis library (YAML), metric catalog and data adapter change. Services were 42.36% of Vietnam’s GDP in 2024.

Anywhere a shift, a queue and a review meet, there’s a “why”.

ROADMAP

1. Real POS adapter (CSV into the same schema)
2. Pilot with one casual-dining restaurant in Ho Chi Minh City, with consent and a PDP Law review
3. Proactive Monday report
4. Hardening: SSO and RBAC, read-only Zetaris account, on-prem or local model lanes through Nightshift

No receipt, no claim.

The challenge: restaurant managers sit on POS, roster and review data, yet still decide on gut feeling when a night goes wrong. With ServiceWhy, a manager asks “why?” and gets findings where every number has a receipt, or an honest “insufficient evidence”. It works by federating four data sources through Zetaris and giving a Verifier the power to block any claim the data cannot reproduce.

Scan for the repository (MIT licence)

github.com/KoangWy/servicewhy